

Customer Satisfaction Analysis of the Olist Marketplace

Data Analytics Report

SQL • Python • Power BI

Portfolio Project

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1. Executive summary

This project investigates the operational factors associated with customer satisfaction in the Olist e-commerce marketplace using SQL, Python, and Power BI.

Key findings include:

- Customer satisfaction was generally high, with **77%** of reviews receiving four or five stars.
- **Delivery performance** exhibited the strongest relationship with customer satisfaction. Orders delivered late received an average review score of **2.27**, compared with **4.29** for on-time deliveries.
- **Freight cost** showed only a weak association with review scores.
- **Seller performance** was broadly consistent across high volume and other sellers, indicating that larger sellers maintained service quality comparable to the rest of the marketplace.

These findings suggest that improving delivery reliability is likely to have the greatest impact on customer satisfaction.

2. Introduction

Background

Olist is a Brazilian e-commerce marketplace that connects customers with independent sellers across Brazil. This project analyzes marketplace transaction data to identify the operational factors associated with customer satisfaction and provide data-driven recommendations for improving customer experience.

Business Problem

Customer satisfaction is a key performance indicator for e-commerce marketplaces because it influences customer retention, seller reputation, and long-term business growth. Understanding the operational factors associated with customer satisfaction enables marketplace operators to improve customer experience and maintain a competitive advantage.

This project investigates which operational factors are most strongly associated with customer satisfaction in the Olist marketplace, with particular emphasis on delivery performance, freight costs, product categories, and seller performance.

Objectives

- Analyze marketplace performance using key business metrics.
- Identify factors associated with customer satisfaction.
- Evaluate the impact of delivery performance, freight costs, product categories, and seller performance on review scores.
- Provide data-driven recommendations to improve customer experience

3. Dataset Overview

The analysis uses the **Brazilian E-Commerce Public Dataset by Olist**, which contains approximately **100,000** customer orders placed between **2016 and 2018**. The dataset captures various aspects of the order lifecycle, including customer purchases, payments, product information, seller information, delivery, and customer reviews. It was obtained from **Kaggle** and is widely used for data analytics and business intelligence projects.

The dataset consists of the following tables:

Table	Description
<i>orders</i>	Order lifecycle
<i>order_items</i>	Products purchased
<i>order_reviews</i>	Customer ratings
<i>order_payments</i>	Payment information
<i>customers</i>	Customer location information
<i>sellers</i>	Seller information
<i>products</i>	Product details
<i>geolocation</i>	ZIP code coordinates
<i>translation</i>	Portuguese to English category names

4. Data cleaning

Issue	Action
Missing review comments	Retained (optional fields)
Missing delivery timestamps	Retained (valid order states)
Missing product information	Retained
Duplicate geolocations	Removed
Misspelled columns	Renamed
Incorrect data types	Converted

No records were removed except duplicate geolocation entries. Missing values corresponding to incomplete orders were retained because they represent valid business events. Date fields were converted to appropriate datetime formats to enable delivery time calculations and monthly trend analysis.

5. Marketplace Overview

Scope of analysis

Since 97% of orders were successfully delivered and customer reviews are only available for completed purchases, the subsequent analyses focus exclusively on delivered orders

5.1. Key Performance Indicators (KPIs)

KPI	Value
Total Delivered Orders	96,478
Total Customers	93,358
Active Sellers	2,970
Gross Merchandise Value (GMV)	R\$15.42 million
Average Order Value	R\$159.68
Average Review Score	4.16 / 5

Note: All monetary values are reported in **Brazilian Reais (BRL)**, the official currency of Brazil.

Key findings

During the analysis period, the Olist marketplace successfully completed **96,478 orders**, serving **93,358 unique customers** through **2,970 active sellers**. These transactions generated approximately **R\$15.42 million in Gross Merchandise Value (GMV)**, with customers spending an average of **R\$159.68 per order**. Overall customer satisfaction was relatively high, with an average review score of **4.16 out of 5**.

5.2. Monthly marketplace activity

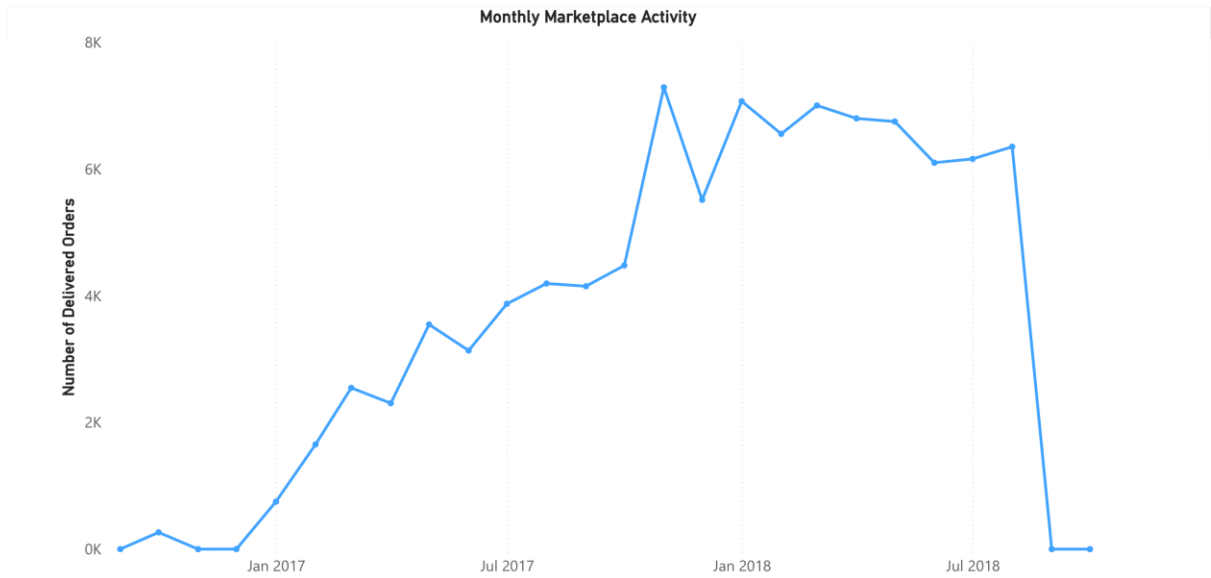


Figure 1. Monthly marketplace activity based on delivered orders (September 2016–October 2018).

Note: The decline observed in September–October 2018 reflects the dataset cutoff rather than a reduction in marketplace activity.

Key Findings

Order purchase records span from September 2016 to October 2018. The monthly trend reveals three distinct phases:

- **September–December 2016:** Marketplace activity was extremely limited, with only a small number of completed orders.
- **January 2017–August 2018:** Delivered orders increased rapidly throughout 2017 before reaching a relatively steady operating level during 2018.
- **September–October 2018:** Delivered orders declined sharply because many recently placed orders had not yet been completed when the dataset was collected.

These patterns provide important context when interpreting the dataset and highlight that the beginning and end of the observation period are not representative of typical marketplace activity.

6. Factors Associated with Customer Satisfaction

6.1. Review score distribution

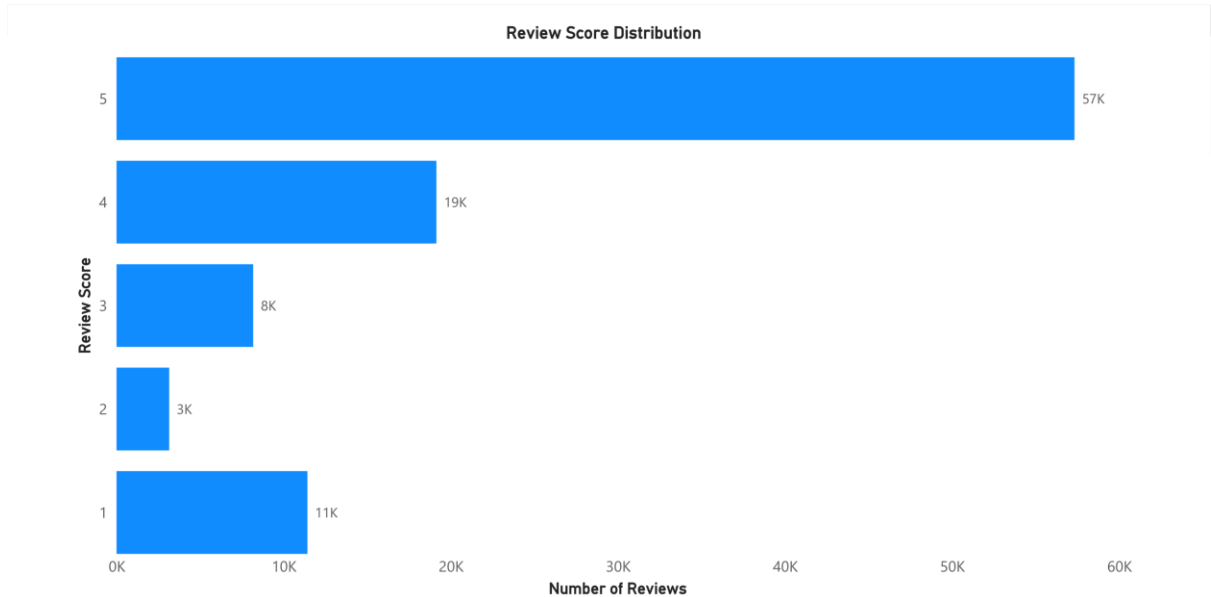


Figure 2. Distribution of Customer Review Scores.

Key findings

Customer ratings are generally positive, with **77.07%** of reviews receiving 4 or 5 stars. In particular, 5-star reviews account for **57.78%** of all ratings, indicating that most customers were satisfied with their purchase experience.

However, **11.51%** of reviews received only 1 star, representing a substantial group of dissatisfied customers. Understanding the factors associated with these low ratings is important for identifying opportunities to improve customer experience.

6.2. Delivery performance

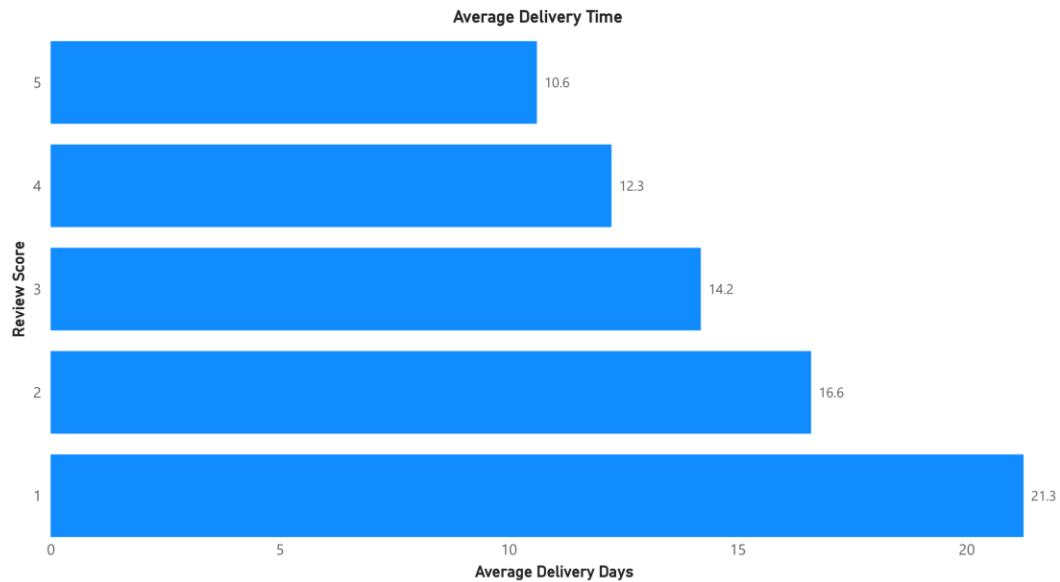


Figure 3. Average Delivery Time (Days) by Customer Review Score.

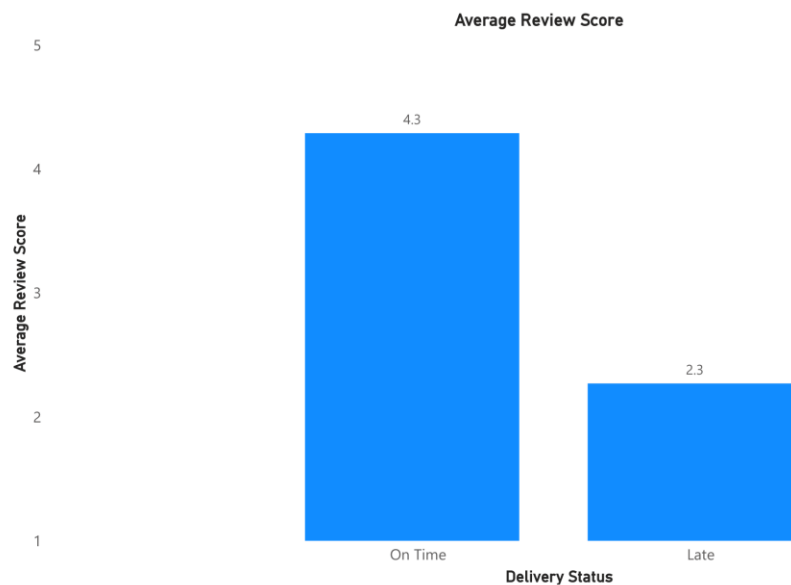


Figure 4. Average Review Score for On-time and Late Deliveries

Key findings

Customers who left 1-star reviews experienced substantially poorer delivery performance than those who left 5-star reviews. On average, 1-star reviews were associated with **delivery times twice as long (21.25 vs. 10.62 days)**. Furthermore, **more than one-third (36.62%)** of 1-star orders were delivered after the estimated delivery date, compared with

only **1.86%** of 5-star orders. These findings suggest that delivery performance is strongly associated with customer satisfaction.

Although only **6.77%** of delivered orders arrived after the promised delivery date, the impact on customer satisfaction was substantial. Orders delivered on time received an average review score of **4.29**, whereas late deliveries averaged only **2.27**, a **difference** of more than **2 review points**.

This finding indicates that **delivery reliability is strongly associated with customer satisfaction**. While late deliveries are relatively uncommon, they correspond to significantly poorer customer experiences and are likely a major contributor to negative reviews.

Business Implication

Maintaining reliable delivery performance appears to be one of the most effective ways to preserve customer satisfaction. Although late deliveries affected fewer than 7% of completed orders, they were associated with an average review score, which was more than two points lower than on-time deliveries. Reducing shipment delays and improving logistics reliability could therefore produce disproportionate improvements in customer experience.

6.3. Freight cost

Review	Average Freight (R\$)	Freight Rate
1	28.16	21.3%
2	26.33	21.8%
3	23.55	21.8%
4	22.35	21.1%
5	21.71	20.6%

Key findings

Orders receiving lower review scores were associated with slightly higher freight charges. However, both the average freight value and freight rate varied only modestly across review groups. Compared with the much stronger relationship observed for delivery performance, shipping cost appears to have a relatively limited association with customer satisfaction.

6.4. Seller performance

Seller Group	Orders	Avg Review	Avg Delivery	% Late
Top 10 Sellers	15,237	4.04	12.45	6.92%
Other Sellers	94,776	4.09	12.36	6.39%

An additional comparison between the ten highest-volume sellers and all remaining sellers revealed only negligible differences in customer ratings, delivery times, and late-delivery rates. This suggests that customer satisfaction is not primarily determined by seller size but is more closely associated with delivery performance across the marketplace.

7. Conclusion

This study investigated the factors associated with customer satisfaction in the Olist marketplace using delivered orders. Following data cleaning and exploratory analysis, the investigation focused on delivery performance, freight cost, and overall marketplace characteristics.

Overall, customer satisfaction was high, with more than three quarters of reviews receiving four or five stars and an average review score of **4.16 out of 5**. Despite this generally positive performance, the analysis identified **delivery reliability** as the factor most strongly associated with customer satisfaction.

Although only **6.77%** of delivered orders arrived after the estimated delivery date, these late deliveries received an average review score of **2.27**, compared with **4.29** for on-time deliveries. This finding suggests that even relatively infrequent delivery delays can have a substantial negative impact on customer experience.

By comparison, freight cost exhibited only a weak relationship with customer satisfaction. Although lower-rated orders incurred slightly higher freight charges, the differences were relatively small and inconsistent. Similarly, comparisons between the marketplace's highest-volume sellers and other sellers revealed only negligible differences in customer ratings and delivery performance, suggesting that seller size alone is not a meaningful indicator of customer satisfaction.

Overall, the findings indicate that **delivery performance is the primary operational factor associated with customer satisfaction in the Olist marketplace**. Improving delivery reliability is therefore likely to produce greater gains in customer satisfaction than reducing shipping charges or focusing on seller size alone.

8. Recommendations

8.1. Prioritize reducing late deliveries

Although fewer than 7% of delivered orders arrived after the promised delivery date, these orders received an average review score of only 2.27, compared with 4.29 for on-time deliveries. Improving logistics reliability should therefore be the highest operational priority.

8.2. Improve operational efficiency rather than reducing freight charges

The analysis found only a weak association between freight cost and customer satisfaction, whereas delivery speed and reliability showed a much stronger relationship. Rather than focusing primarily on reducing shipping prices, Olist should prioritize initiatives that improve fulfillment and delivery performance.

8.3. Monitor delivery performance using operational KPIs

Regularly tracking metrics such as average delivery time, percentage of delayed orders, and customer review scores can help identify deteriorating logistics performance early and enable timely corrective actions.